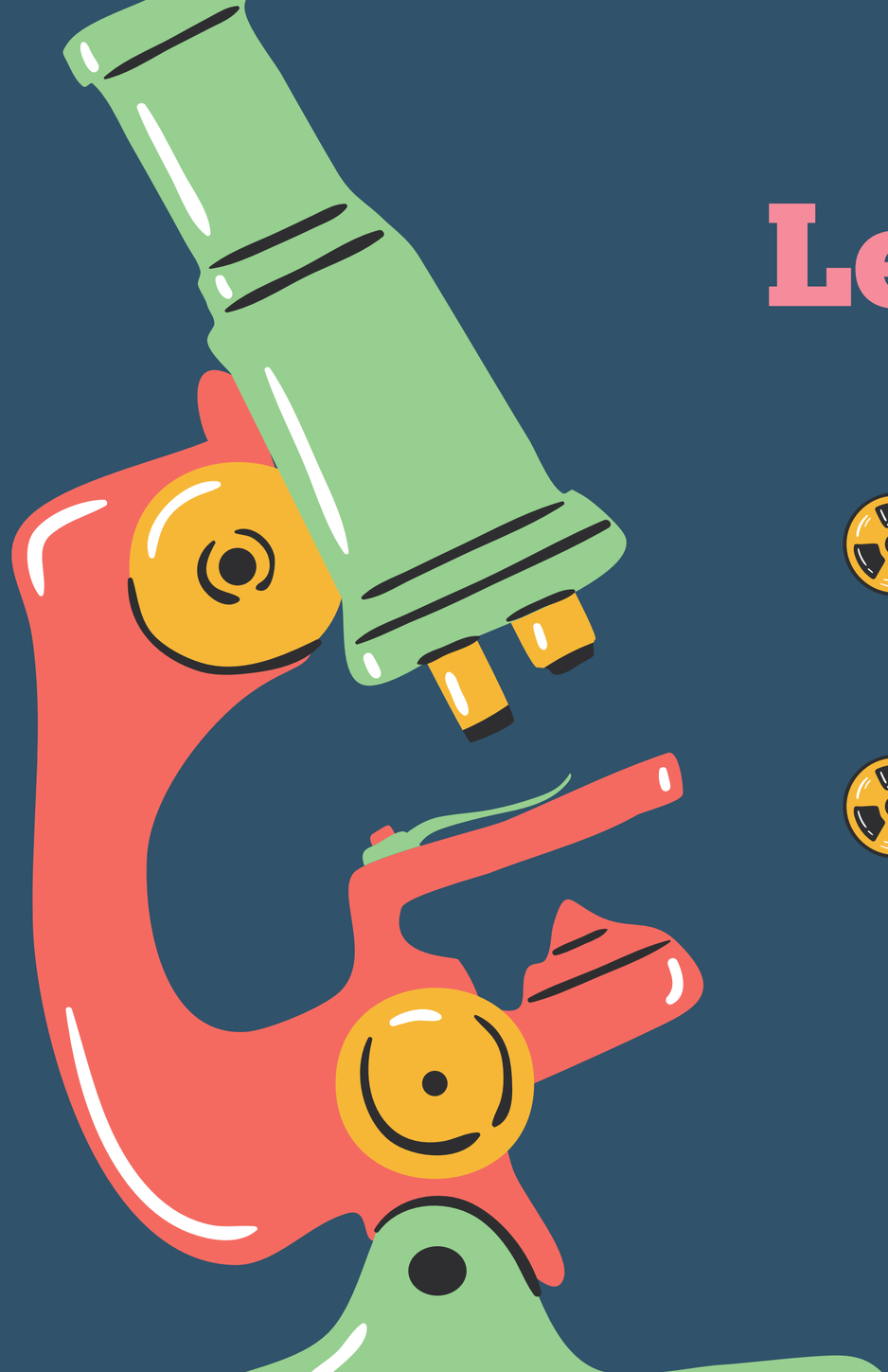


Chemical Names and Formulas

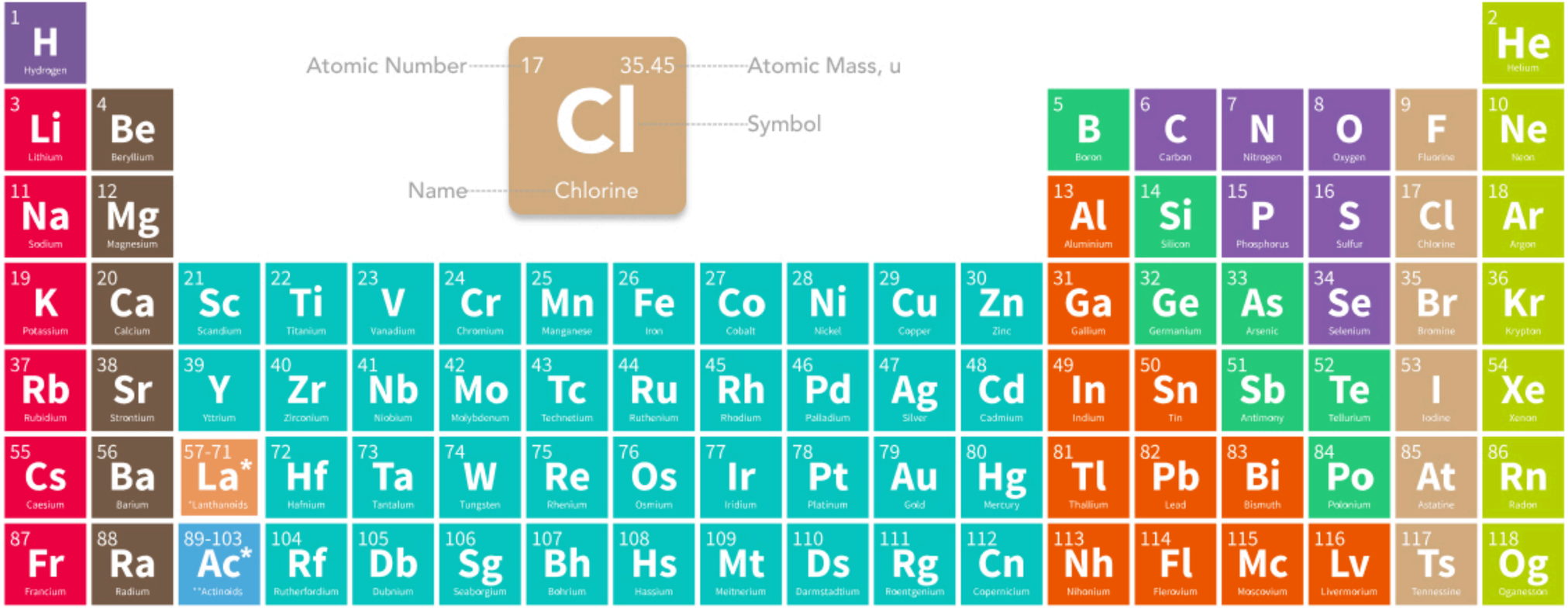


Learning Objectives:

- Recognize the formulas of common chemical substances
- Name some common chemical substances given their formula and write formulas given the names of some common chemical substances.

STANFORD ADVANCED MATERIALS

PERIODIC TABLE OF ELEMENTS



*Lanthanoids	57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
*Actinoids	89 Ac Actinium	90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

The Modern Periodic Table

- hydrogen
- alkali metals
- alkali earth metals
- transition metals
- Poor metals
- nonmetals
- noble gases
- rare earth metals

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H Hydrogen 1.008																	2 He Helium 4.003
3 Li Lithium 6.941	4 Be Beryllium 9.012											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453	18 Ar Argon 39.948
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 84.798
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.294
55 Cs Cesium 132.905	56 Ba Barium 137.328	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [208.982]	85 At Astatine 209.987	86 Rn Radon 222.018
87 Fr Francium 223.020	88 Ra Radium 226.025	89-103	104 Rf Rutherfordium [262]	105 Db Dubnium [261]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [278]	110 Ds Darmstadtium [281]	111 Rg Roentgenium [280]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [289]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]

57 La Lanthanum 138.905	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.243	61 Pm Promethium 144.913	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.925	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967
89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium 257.095	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr Lawrencium [262]

COMPOUND

Any substance composed of identical molecules consisting of atoms of two or more chemical elements.



CHEMICAL FORMULA

Identifies each constituent element by its chemical symbol and indicated the proportionate number of atoms of each element.



STRUCTURAL FORMULA

A formula which shows the arrangement of atoms in the molecule of a compound.



LEWIS DOT STRUCTURE

Is a visual representations of the valence electrons in atoms and molecules.



IONIC COMPOUNDS

- Consists of Cations (positive ions) and Anions (negative ions) held together by electrostatic attraction.
- transferring of electrons
- Usually metal and non-metal
- monoatomic ions and polyatomic ions

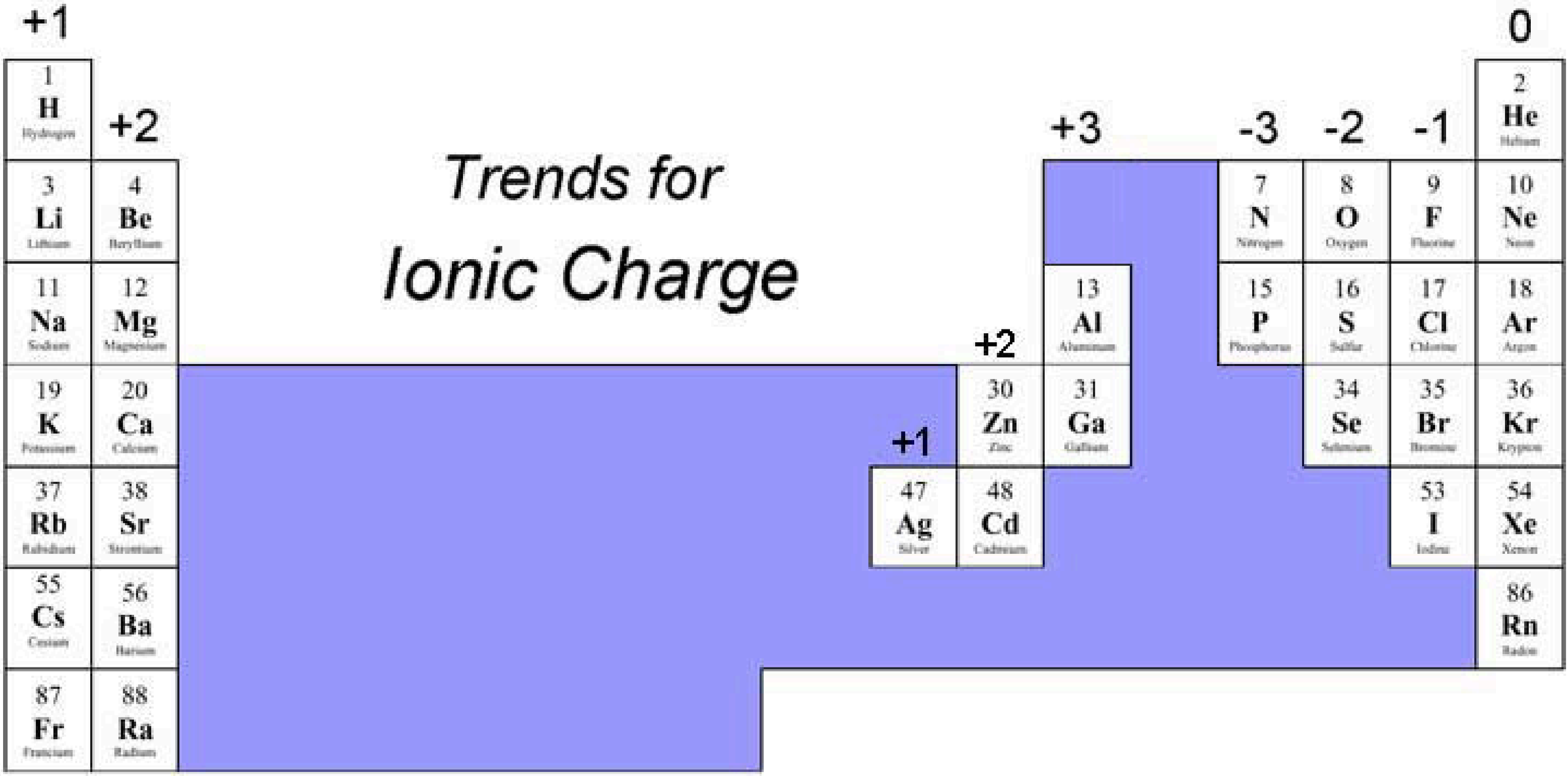


MOLECULAR COMPOUNDS

- Consists of non-metal atoms held together by covalent bonds.
- Sharing of electrons
- Covalent or molecular compounds contain atoms held together by covalent bonds.



Trends for Ionic Charge



MONOATOMIC CATION

"Mono" means one.

"Atomic" refers to atom.

A monoatomic ion is a single atom that has gained or lost electrons, resulting in a charge.

Cation (positive charge) – the atom loses electrons

- Sodium ion: Na^+ (sodium loses 1 electron)
- Calcium ion: Ca^{2+} (calcium loses 2 electrons)

Anion (negative charge) – the atom gains electrons

- Chloride ion: Cl^- (chlorine gains 1 electron)
- Oxide ion: O^{2-} (oxygen gains 2 electrons)

Group IA elements form +1 ions: Li^+ , Na^+ , K^+ , Rb^+ , Cs^+
Group IIA elements form +2 ions: Be^{+2}, Mg^{+2}, Ca^{+2}, Sr^{+2}, Ba^{+2}

silver ion = Ag^{+1}

cadmium ion = Cd^{+2}

aluminum ion = Al^{+3}

zinc ion = Zn^{+2}

HOW WE SHOULD NAME THEM?

element name + ion

MONOATOMIC CATION

1. Sr^{+2}
2. Ga^{+3}
3. Ag^{+1}
4. Ba^{+2}
5. Cd^{+2}

TRANSITION METALS

The Stock system is used to name transition metals, Sn, and Pb that each form more than one ion:

element name (charge in Roman numerals) + ion

Fe^{+2} = _____ Cu^{+} = _____

Fe^{+3} = _____ Cu^{+2} = _____

TRANSITION METALS

1. Cr^{+2}
2. Ti^{+4}
3. Pd^{+2}
4. Sc^{+3}
5. Pt^{+4}
6. Hg^{+2}
7. Sn^{+4}
8. V^{+5}
9. Pb^{+2}
10. Ni^{+3}